

Quantum Math

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Basis Algebra

Linear Combinations

A vector $\vec{v}$ can be expressed as:

$$\vec{v} = c_1 \vec{v}_1 + c_2 \vec{v}_2 + \dots + c_n \vec{v}_n$$

where c_i are scalar coefficients.

Span of a Set

The **span** of vectors $\{v_1, v_2, \dots, v_n\}$ is the set of all possible linear combinations:

$$\text{span}(S) = \left\{ \sum_{i=1}^n c_i v_i : c_i \in C \right\}$$

Linear Independence

Vectors are **linearly independent** if:

$$c_1 v_1 + c_2 v_2 + \dots + c_n v_n = 0 \implies c_1 = c_2 = \dots = c_n = 0$$

Basis Requirements:

1. Vectors must be linearly independent
2. Vectors must span the entire space
3. For n -dimensional space, need exactly n vectors

Change of Basis

Given two bases $B_1 = \{u_1, u_2, \dots, u_n\}$ and $B_2 = \{v_1, v_2, \dots, v_n\}$:

Transformation Matrix

If $v_j = \sum_{i=1}^n P_{ij} u_i$, then the transformation matrix is:

$$P = \begin{pmatrix} P_{11} & P_{12} & \dots & P_{1n} \\ P_{21} & P_{22} & \dots & P_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ P_{n1} & P_{n2} & \dots & P_{nn} \end{pmatrix}$$

Vector Transformation

$$[\vec{v}]_{B_2} = P^{-1} [\vec{v}]_{B_1}$$

Quantum Computational Basis States

Standard Basis States:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (1)$$

$$|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (2)$$

General Qubit State

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

where $|\alpha|^2 + |\beta|^2 = 1$ (normalization condition).

Multi-Qubit Systems

For n qubits, the computational basis consists of 2^n states:

$$|i\rangle = |i_{n-1} i_{n-2} \dots i_1 i_0\rangle$$

where $i = \sum_{j=0}^{n-1} i_j 2^j$ and $i_j \in \{0, 1\}$.

Hilbert Space

Mathematical Structure Hierarchy

1. **Topological Space:** Set with open sets
2. **Metric Space:** Distance function $d(x, y)$
3. **Vector Space:** Linear operations
4. **Normed Space:** Norm function $\|\cdot\|$
5. **Banach Space:** Complete normed space
6. **Inner Product Space:** Inner product $\langle \cdot, \cdot \rangle$
7. **Hilbert Space:** Complete inner product space

Inner Product Properties:

1. $\langle x, x \rangle \geq 0$ (positive definiteness)
2. $\langle x, x \rangle = 0 \iff x = 0$
3. $\langle x, y \rangle = \overline{\langle y, x \rangle}$ (conjugate symmetry)
4. $\langle ax + by, z \rangle = a \langle x, z \rangle + b \langle y, z \rangle$ (linearity)

Induced Norm

$$\|x\| = \sqrt{\langle x, x \rangle}$$

Quantum Algebra Operations

Basic Operations

- **Addition:** $|\psi_1\rangle + |\psi_2\rangle$
- **Scalar Multiplication:** $c|\psi\rangle$
- **Inner Product:** $\langle\phi|\psi\rangle$
- **Outer Product:** $|\psi\rangle\langle\phi|$

Tensor Products

For multi-particle systems:

$$|\psi\rangle \otimes |\phi\rangle = |\psi, \phi\rangle = |\psi\rangle |\phi\rangle$$

Measurement Probabilities

Probability of measuring state $|\phi\rangle$ in system $|\psi\rangle$:

$$P = |\langle\phi|\psi\rangle|^2$$

Matrix Representation

State Vectors

Quantum states are represented as column vectors:

$$|\psi\rangle = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix}$$

Operators

Quantum operators are represented as matrices:

$$\hat{A} = \begin{pmatrix} A_{11} & A_{12} & \cdots & A_{1n} \\ A_{21} & A_{22} & \cdots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{n1} & A_{n2} & \cdots & A_{nn} \end{pmatrix}$$

Matrix Elements

$$A_{ij} = \langle i | \hat{A} | j \rangle$$

Important Matrix Properties:

- **Hermitian:** $A^\dagger = A$ (observable quantities)
- **Unitary:** $U^\dagger U = I$ (quantum gates)
- **Trace:** $\text{Tr}(A) = \sum_i A_{ii}$
- **Determinant:** $\det(A)$ (matrix invertibility)

Key Quantum Mathematical Identities

Completeness Relation

$$\sum_i |i\rangle\langle i| = I$$

Resolution of Identity

For orthonormal basis $\{|i\rangle\}$:

$$|\psi\rangle = \sum_i \langle i|\psi\rangle |i\rangle$$

Expectation Value

$$\langle\hat{A}\rangle = \langle\psi|\hat{A}|\psi\rangle$$

Uncertainty Relation

$$\Delta A \Delta B \geq \frac{1}{2} |\langle[\hat{A}, \hat{B}]\rangle|$$

where $[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$ is the commutator.